# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**
**COURSE CODE: CSA07**

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**COURSE OUTCOME COVERED**

***CO5:*** Measure network performance using key metrics with simulation tools. *(BL4,BL5)*

*Assessment Tool Weightage: 30%*

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Annexure A

# Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the

bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance.Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

## RUBRICS FOR CALCULATION

| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Identifies all inputs, outputs, | Minor missing elements | Partial understanding | Incorrect interpretation |

| | | | | | |
|---|---|---|---|---|---|
| | | constraints accurately | | | |
| Algorithm Design | 30% | Complete, correct, and logically sound solution | Minor logical gaps | Partially correct logic | Incorrect approach |
| Use of Control Structures | 20% | Efficient and appropriate use of loops, conditions, functions/modules | Minor inefficiencies | Limited or basic usage | Incorrect or missing constructs |
| Pseudocode Structure | 10% | Well-structured, clear, consistent format, easy to follow | Mostly clear with minor issues | Difficult to follow in parts | Poorly structured and unclear |
| Completeness | 20% | Handles all cases; optimized approach | Handles most cases; reasonable efficiency | Handles basic cases only | Incomplete or inefficient |

# 1. Rural Branch Office – Internet Connectivity and Security

### Problem Analysis

The branch office has:

- No high-speed wired Internet.
- Need for cloud applications and video conferencing.
- Need for secure communication with headquarters.

- Limited connectivity budget.
- Requirement for uninterrupted business operations.
- Need for future expansion.

## Recommended Network Design

```
   Satellite / 4G/5G Internet
            |
     Dual-WAN Router
            |
      Firewall/UTM
            |
     Managed Switch
   /      |      \
 PCs   Printer   Wi-Fi AP
            |
     VPN to HQ
```

## Recommended Components

| Component | Recommendation | Reason |
|---|---|---|
| Primary Internet | 4G/5G fixed wireless | Lower deployment cost than wired infrastructure where cellular coverage is available |
| Backup Internet | Satellite or second cellular provider | Provides redundancy |
| Router | Dual-WAN business router | Automatically switches between connections |
| Firewall | Next-generation firewall/UTM | Protects the branch from Internet threats |
| Switch | Managed Gigabit switch | VLANs and network management |
| Wi-Fi | Business-grade Wi-Fi 6 AP | Supports employees and mobile devices |
| HQ connection | Site-to-site VPN | Encrypts branch-to-HQ traffic |
| UPS | UPS for networking equipment | Maintains operation during short power interruptions |

## Bandwidth Strategy

Because bandwidth may be limited, the company should prioritize business traffic:

1. Cloud applications and VPN traffic get high priority.
2. Video conferencing gets sufficient QoS priority.
3. Non-business downloads and streaming are restricted.
4. Usage monitoring helps identify bandwidth problems.

## Reliability

A **dual-WAN configuration** is recommended. The primary connection can be 4G/5G, while a secondary provider or satellite connection acts as backup.

If the primary connection fails:

```
Primary Internet ──X──> Router
                  |
            Backup Internet
                  |
                Office
```

Traffic can automatically move to the backup connection.

## Security

- Site-to-site **IPsec VPN** between branch and headquarters.
- Next-generation firewall.
- WPA3 Enterprise for wireless access.
- Strong administrator authentication.
- Regular firmware and security updates.
- Separate guest Wi-Fi from company devices.
- Endpoint protection on employee computers.

## Scalability

The managed switch should have additional ports, and the router/firewall should support increased bandwidth and additional VPN connections.

## Justification

A **4G/5G primary connection with a backup connection, dual-WAN router, firewall, and VPN** provides a reasonable balance between cost, reliability, security, and performance. It avoids waiting for expensive wired infrastructure while allowing the branch to expand later.

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# 2. University Wireless Network – 3,000+ Students

## Problem Analysis

The university requires:

- More than 3,000 simultaneous users.
- Coverage across classrooms, laboratories, libraries, and hostels.
- Limited number of access points.
- Low interference.
- Secure authentication.
- Good performance at reasonable cost.

The key challenge is that **coverage alone is not enough**. High-density areas need sufficient wireless capacity.

## Proposed Architecture

```
             Internet
                |
           Core Firewall
                |
          Core Switches
                |
      ---------------------
      |         |         |
  Building A Building B Building C
      |         |         |
   PoE Switches / Access Switches
      |         |         |
     APs       APs       APs
      |
  Students / Staff
```

## Access Point Placement

APs should be placed after a **wireless site survey**, rather than simply placing one AP per building or floor.

Priority should be given to:

- Classrooms with high student density.

- Lecture halls.
- Libraries.
- Laboratories.
- Hostel common areas.
- Corridors and outdoor student areas where required.

High-density locations should receive more AP capacity than low-usage areas.

## Frequency Selection

| Band | Use |
|---|---|
| 2.4 GHz | Longer range and compatibility with older devices |
| 5 GHz | Main band for high-performance student access |
| 6 GHz | Use where supported by Wi-Fi 6E/7 clients and infrastructure |

The design should avoid excessive use of 2.4 GHz because it has fewer non-overlapping channels and is more susceptible to interference.

## Channel Planning

Instead of allowing every AP to use the same channel:

AP1 → Channel A
AP2 → Channel B
AP3 → Channel C
AP4 → Channel A
AP5 → Channel B

A centrally managed wireless controller can automatically optimize channels and transmit power.

## Authentication and Security

For students and staff:

- **WPA2/WPA3-Enterprise**
- 802.1X authentication
- RADIUS authentication server
- Individual student/staff credentials

For guests:

- Separate guest SSID.
- Internet-only access.
- Client isolation where appropriate.

### VLAN Design

Student Wi-Fi → Student VLAN
Staff Wi-Fi → Staff VLAN
Guest Wi-Fi → Guest VLAN
IoT Devices → IoT VLAN

This prevents unnecessary communication between different user groups.

### Balancing Coverage, Performance, Security and Cost

| Requirement | Solution |
|---|---|
| Coverage | Strategic AP placement |
| High density | Additional APs in crowded areas |
| Interference | Channel and power management |
| Performance | Prefer 5 GHz/6 GHz |
| Security | WPA3/802.1X/RADIUS |
| Cost | APs concentrated where demand is highest |
| Scalability | Central wireless management |

### Justification

With a limited AP budget, the university should **not attempt to maximize physical coverage at the expense of capacity**. APs should be concentrated in high-density locations, with centrally managed channel allocation and transmit power. This provides better performance for 3,000+ users while avoiding unnecessary AP deployment.

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# 3. Financial Institution – Secure Network Architecture

## Problem Analysis

The financial institution requires:

- Strong cybersecurity.
- Protection of sensitive customer information.
- Access to online banking services.
- Strict security and compliance requirements.
- High network performance.
- Controlled operational costs.

## Proposed Architecture

```
               Internet
                  |
         Perimeter Firewall
            /          \
      VPN Gateway    DMZ
                       |
               Web/Application
                  Servers
                       |
               Internal Firewall
                       |
          -----------------------
          |         |          |
      User VLAN   Server VLAN   Admin VLAN
          |         |
      Employees   Database
```

## Network Segmentation

The network should be divided into separate security zones.

| Zone | Purpose |
|---|---|
| User VLAN | Employee computers |
| Server VLAN | Internal application servers |
| Database VLAN | Sensitive customer databases |
| Management VLAN | Network administration |
| Guest VLAN | Visitor Internet access |

| | |
|---|---|
| DMZ | Public-facing services |

The **database VLAN** should have the strongest restrictions.

For example:

Employee → Application Server → Database
✓ ✓ ✓

Employee → Database directly
✗

This reduces the possibility of unauthorized access.

## Firewall Deployment

Use multiple firewall/security layers:

1. **Perimeter firewall** protects the organization from Internet traffic.
2. **Internal segmentation firewall** controls traffic between sensitive network zones.
3. **Web Application Firewall (WAF)** protects public-facing web applications where required.

Firewall rules should follow the **least-privilege principle**: only necessary traffic is allowed.

## Authentication

For employees accessing sensitive systems:

- Multi-factor authentication (MFA).
- Strong password policies.
- Role-based access control.
- 802.1X for network access.
- Privileged access management for administrators.

Example:

Employee
↓
Username + Password
↓
MFA Verification
↓
Role Verification
↓

Authorized Application

## Intrusion Detection and Prevention

Deploy **IDS/IPS** to monitor suspicious traffic.

The system can identify:

- Port scanning.
- Malware traffic.
- Unauthorized access attempts.
- Suspicious network behavior.
- Known attack signatures.

Security logs should be centralized for monitoring and investigation.

## Performance Considerations

Security controls should be placed strategically rather than inspecting unnecessary traffic repeatedly.

- Use hardware-accelerated firewalls where appropriate.
- Apply QoS to critical applications.
- Use VLANs to reduce unnecessary broadcast traffic.
- Monitor firewall and IDS/IPS performance.
- Use redundant security devices for critical services.

## Cost Control

Instead of deploying expensive security appliances everywhere:

- Concentrate advanced security controls around critical systems.
- Use VLANs and access-control policies for logical segmentation.
- Centralize logging and monitoring.
- Automate security alerts and routine administration.
- Use existing managed switches where they meet security requirements.

## Compliance

The institution should map its security controls to the specific regulations and standards applicable to its jurisdiction and banking operations. Security policies should include:

- Access control.
- Data protection.
- Audit logging.

- Incident response.
- Regular vulnerability assessment.
- Backup and recovery.
- Periodic security audits.

### Justification

The proposed architecture uses **segmentation, layered firewalls, MFA, role-based access, and IDS/IPS** to protect sensitive financial information. At the same time, VLANs, centralized management, and selective inspection help control operational costs and avoid unnecessary performance overhead.

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# Overall Comparison

| Requirement | Rural Branch | University | Financial Institution |
|---|---|---|---|
| Main challenge | Limited Internet availability | High-density wireless users | Security of sensitive data |
| Primary solution | 4G/5G + backup | Managed high-density Wi-Fi | Segmented secure architecture |
| Reliability | Dual-WAN | Redundant network infrastructure | Redundant security devices |
| Security | Firewall + VPN | WPA3/802.1X | MFA + firewalls + IDS/IPS |
| Scalability | Additional WAN/LAN capacity | Add APs based on demand | Additional VLANs/security zones |
| Cost strategy | Wireless instead of expensive wired deployment | APs concentrated by demand | Layered security with centralized management |

## Final Conclusion

The three scenarios require **different network strategies because their constraints are different**.

- The **rural branch** should prioritize affordable connectivity and redundancy using **4G/5G, backup connectivity, dual-WAN, firewall, and VPN**.

- The **university** should prioritize wireless capacity using **strategic AP placement, 5/6 GHz, channel management, and WPA2/WPA3-Enterprise with 802.1X**.
- The **financial institution** should prioritize security through **network segmentation, layered firewalls, MFA, RBAC, and IDS/IPS**.

Thus, the best network design is not simply the most expensive one; it is the design that **satisfies the required performance, reliability, security, scalability, and budget constraints**.